
Youping Sun

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Address: 1725 Orrington Ave, Evanston, IL

EDUCATION

Shandong University(SDU), Jinan, China

Sept 2022 – Jun 2026

BEng in Robotics Engineering, School of Control Science and Engineering

- GPA: 84.74/100

University of Wisconsin–Madison (WISC), Madison, WI, United States

Sept 2025 – Dec 2025

Exchange Program, Department of Industrial and Systems Engineering

- GPA: 4.0/4.0

PUBLICATION

- **Y. Sun**, J. Wang, Z. Wang, J. Li, Y. Li, R. Song, and H. Zhang, “Gait Phase Detection and Prediction with Machine Learning Models Based on sEMG,” IEEE AIHCIR, 2024.
- **Y. Sun**, X. Li, C. Ma, X. Ma, and H. Zhang, “Exploring the Correlation Between Level Walking and Stair Ambulation for Fine-Grained Gait Phase Prediction,” ICCA 2026.
- **Y. Sun**, H. Wan, X. Li, F. Zhang, W. Wu, and H. Zhang, “Cross-Locomotion Task Generalization in sEMG-Based Gait Phase Prediction Using Artificial Neural Networks,” AIACT 2026, pp. 77–81.

RESEARCH

The Flexible Electronics and Additive Printing Lab, UW–Madison

Sept 2025 – Dec 2025

Research Assistant, Supervisor: Prof. Hantang Qin

Project: EHD Printing — Model Construction & Scanning, Pattern 3D Projection and CFD Simulation

- Designed and fabricated 3D models with continuously varying curvature in SolidWorks; performed high-precision 3D scanning (0.10 mm accuracy) to assess dimensional fidelity.
- Projected 2D patterns onto irregular surfaces in Blender and resampled toolpaths to 1 mm point spacing for fabrication.
- Built a COMSOL Multiphysics CFD/electrostatics model to optimise electric-field distribution in EHD printing, improving agreement between simulations and experimental data.

The Center for Robotics, School of Control Science and Engineering, SDU

Oct 2024 – Sept 2026

*Research Assistant, Supervisor: **Prof. Huanghe Zhang***

Project: Gait Phase Detection and Prediction with Machine Learning Models

- Analysed sEMG signals from human lower-limb movements and predicted gait phases using machine learning models.
- Extracted feature values from raw data to train ANN and CNN models, achieving higher detection and prediction accuracy compared with KNN and SVM baselines.

PROJECTS

The 19th “Challenge Cup” National Student Innovation Contest

July 2025 – Nov 2025

- Developed robot control code in a Docker-based simulation environment, enabling the humanoid robot in

a warehouse to autonomously pick items on a table, navigate to the shelves and place them in right place.

Intelligent Quadruped Robot (*Intelligent Robot Practice*)

Feb 2025 – Mar 2025

- Conducted simulation experiments in Webots on Ubuntu, implementing pose transformation and motion control for robotic dog line-following tasks. Achieved obstacle avoidance at high speed by using data captured from LiDAR and cameras.

The 19th National College Student Smart Car Race

Sept 2023 – Aug 2024

- Implemented track recognition using OpenCV-based image processing, distinguishing curve types for adaptive control.
- Achieved high-speed car motion via PID control algorithms integrated with ROS.

HONORS & AWARDS

- **Second Prize**, “AI+” Special Track, 19th “Challenge Cup” **National** Student Innovation Contest, 2025
- **First Prize**, The 19th **National** College Student Smart Car Race, 2024
- **Outstanding Individual**, Student Innovation and Entrepreneurship Activities, SDU, 2023
- **Entrepreneurship Practice Scholarship**, SDU, 2023
- **Outstanding Individual in Social Practice**, SDU, 2023

SKILLS

Programming: C/C++, Python, R, Matlab

Mathematics: Calculus, Probability, Linear Algebra